

SOC INVESTIGATION REPORT

Threat Detection Using Splunk SIEM

Brute Force | Credential Theft | Ransomware | Insider Threat

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Education	MSc Cybersecurity — Nottingham Trent University, UK
Experience	1 Year — Consulting Engineer, Cisco
Tool Used	Splunk SIEM — Search Processing Language (SPL)
Log Files	windows_security.log investigation3.log
Date	June 27, 2026
Severity	CRITICAL

1. Executive Summary

On June 25 and 26 2026, two separate but critical security incidents were detected and investigated using Splunk SIEM. The first incident involved an external attacker who successfully brute forced the admin account on DESKTOP-CQANST8, stole credentials using mimikatz, created a backdoor account, established persistence through a scheduled task, and cleared audit logs to cover their tracks. The second incident involved a more complex attack combining an external RDP brute force, ransomware deployment that encrypted 5 critical files, and a simultaneous insider threat where a legitimate employee was caught exfiltrating sensitive company data to an external IP via FTP.

Both incidents are rated Critical severity and require immediate containment action. The full attack chains, evidence, MITRE ATT&CK mappings, and containment steps are documented in this report.

Incidents 2	Severity CRITICAL	Attacker IPs 3 Identified	Files Encrypted 5 Files
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2. Tools and Scope

SIEM Tool	Splunk Enterprise 10.4.0
Query Language	SPL — Search Processing Language
Log File 1	windows_security.log — 30 events — June 25 2026
Log File 2	investigation3.log — 43 events — June 26 2026
Target Machine	DESKTOP-CQANST8
Investigation Period	June 25 2026 08:00 AM to June 26 2026 09:00 AM

3. Investigation 1 — Brute Force and Credential Theft Attack

This investigation covers a targeted attack against the admin account on DESKTOP-CQANST8. The attacker came from an external IP address and used a brute force technique to gain access via the network. Once inside, they deployed credential stealing malware, created a backdoor account, set up persistence, and cleared all evidence before leaving.

3.1 All Events Overview

The initial search across all 30 events in windows_security.log gave a complete picture of activity on the machine. The timeline showed normal employee logins in the morning followed by a cluster of suspicious activity starting at 08:10 AM.

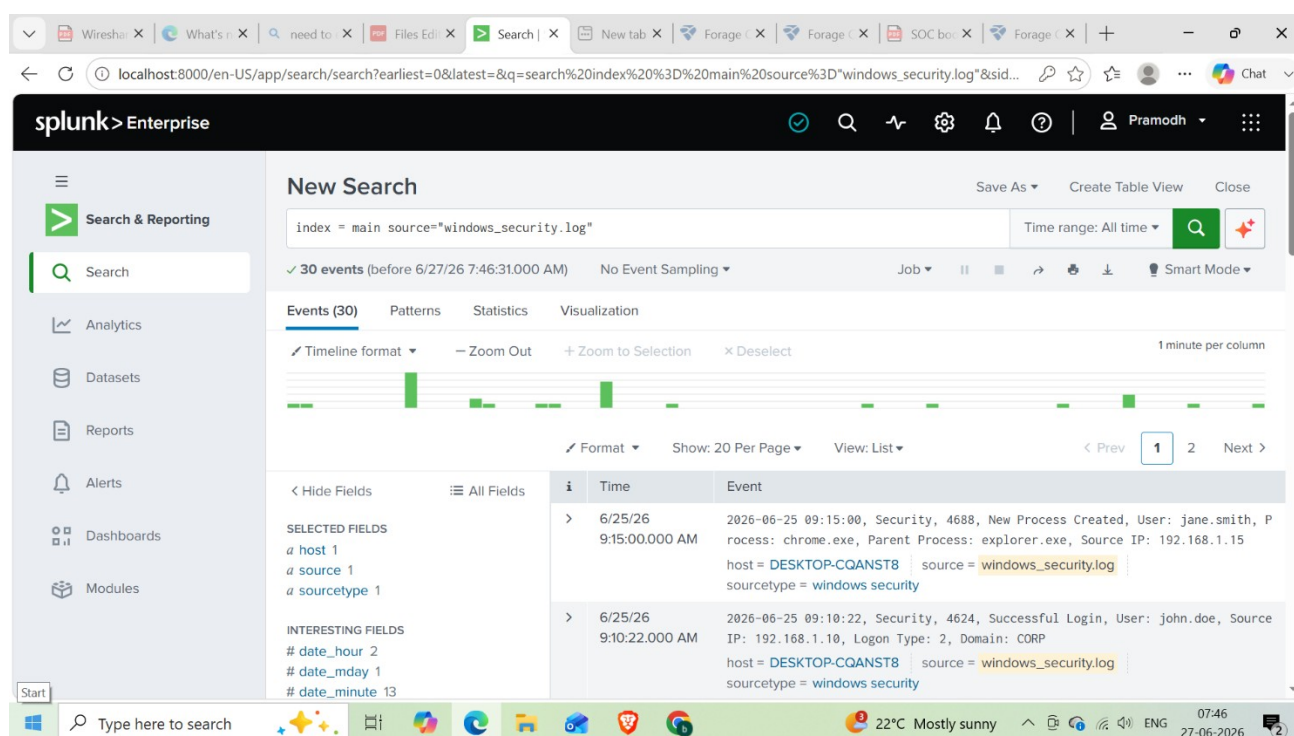


Figure 1 — All 30 events loaded from windows_security.log in Splunk

3.2 Brute Force Detection — Failed Logins

Filtering for failed login events revealed 15 events across the log file. Three separate source IPs were targeting three different user accounts on the same machine. The most significant cluster was 7 consecutive failed logins against the admin account from IP 192.168.1.50 between 08:10:01 and 08:10:13, all within 12 seconds. This is a clear brute force pattern — an automated tool trying multiple passwords in rapid succession.

- **IP 192.168.1.50:** 7 failed attempts on admin account using Logon Type 3 which is Network
- **IP 10.0.0.99:** 5 failed attempts on administrator account using Logon Type 10 called RDP
- **IP 172.16.0.200:** 3 failed attempts on john.doe account using Logon Type 3 which is Network

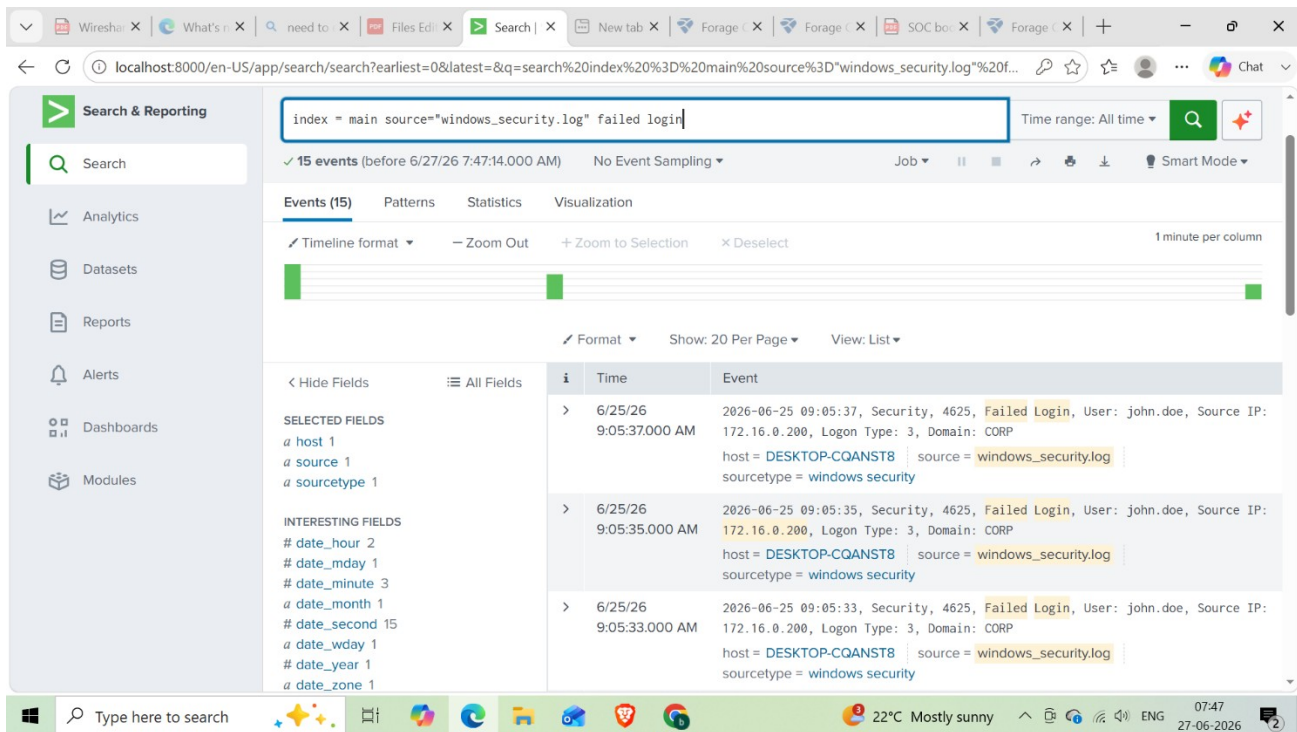


Figure 2 — 15 failed login events showing brute force attempts from multiple IPs

3.3 Successful Login — Brute Force Succeeded

After 7 failed attempts, the attacker from IP 192.168.1.50 successfully logged into the admin account at exactly 08:10:15 using Logon Type 3 a network login. This is the moment the breach occurred. The time gap between the last failed attempt and the successful login was just 2 seconds, confirming this was an automated brute force tool that found the correct password.

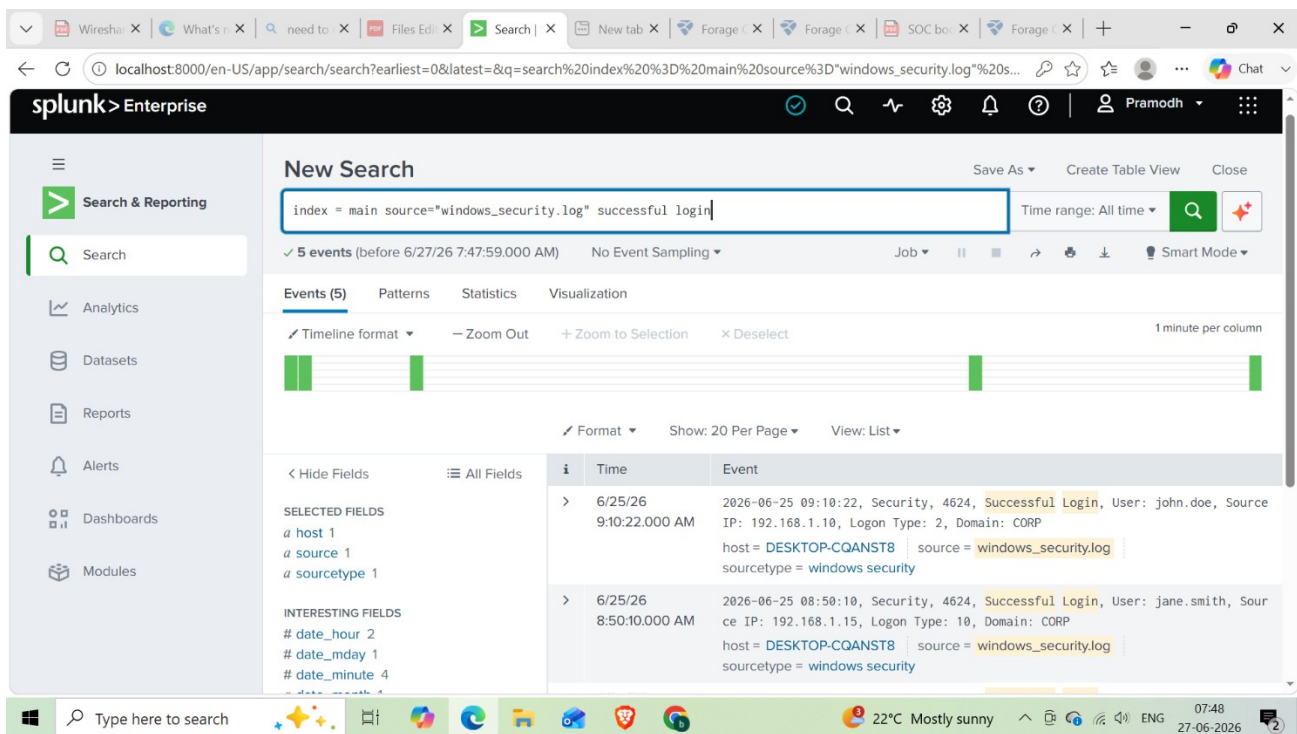


Figure 3 — Successful login by attacker from 192.168.1.50 after brute force

3.4 Credential Theft — Mimikatz Execution

At 08:16:10, just 6 minutes after gaining access, the attacker launched mimikatz.exe. Mimikatz is a well-known credential dumping tool that extracts password hashes directly from Windows memory. The parent process was powershell.exe, which was itself launched from cmd.exe a classic attacker process chain showing the attacker set up their environment before running the credential theft tool. With the stolen hashes, the attacker can now authenticate to other machines on the network without needing to know the actual passwords a technique called Pass the Hash.

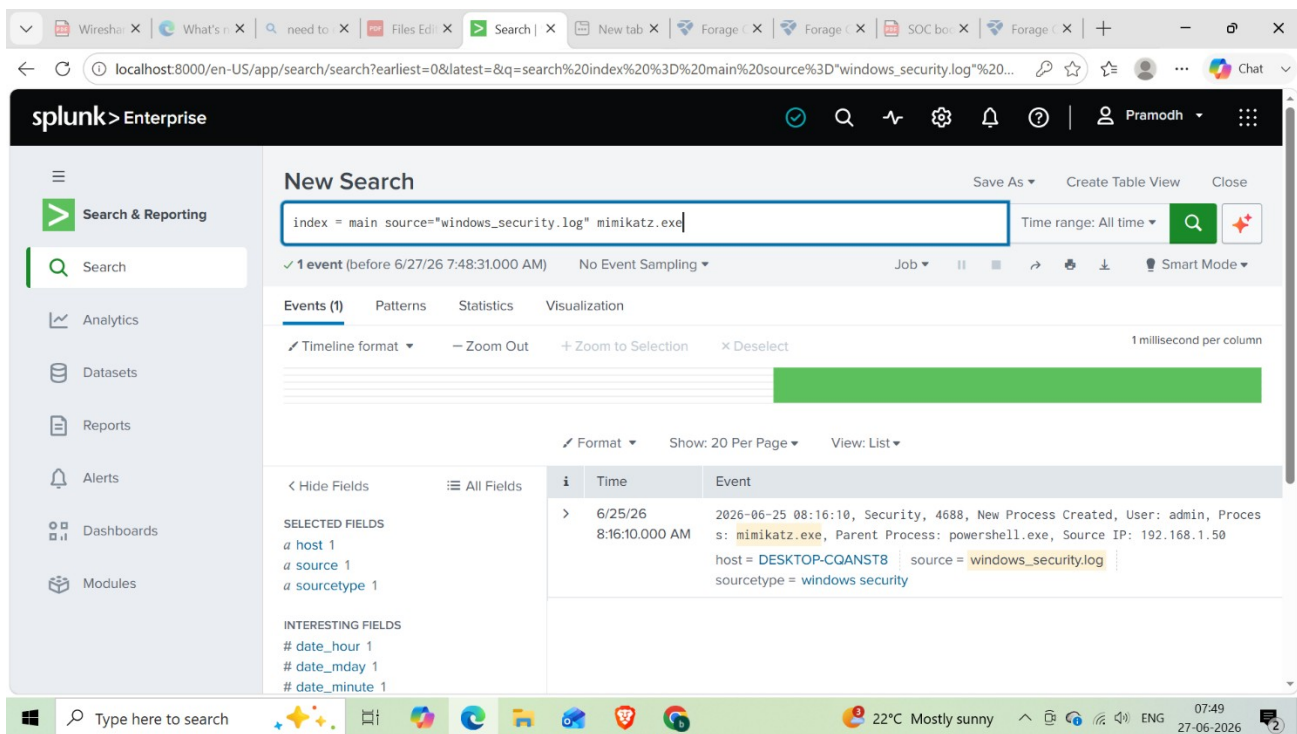


Figure 4 — Mimikatz executed by admin user from attacker IP — credential dumping confirmed

3.5 Persistence — Backdoor Account Created

At 08:20:33, the attacker created a new user account called `backdoor_user` using Event ID 4720. One minute later at 08:21:00, they immediately added this account to the Administrators group using Event ID 4728. This gives the attacker a permanent way back into the machine even if the admin password is changed, the `backdoor_user` account still exists and has full admin privileges. This is called persistence in attack terminology.

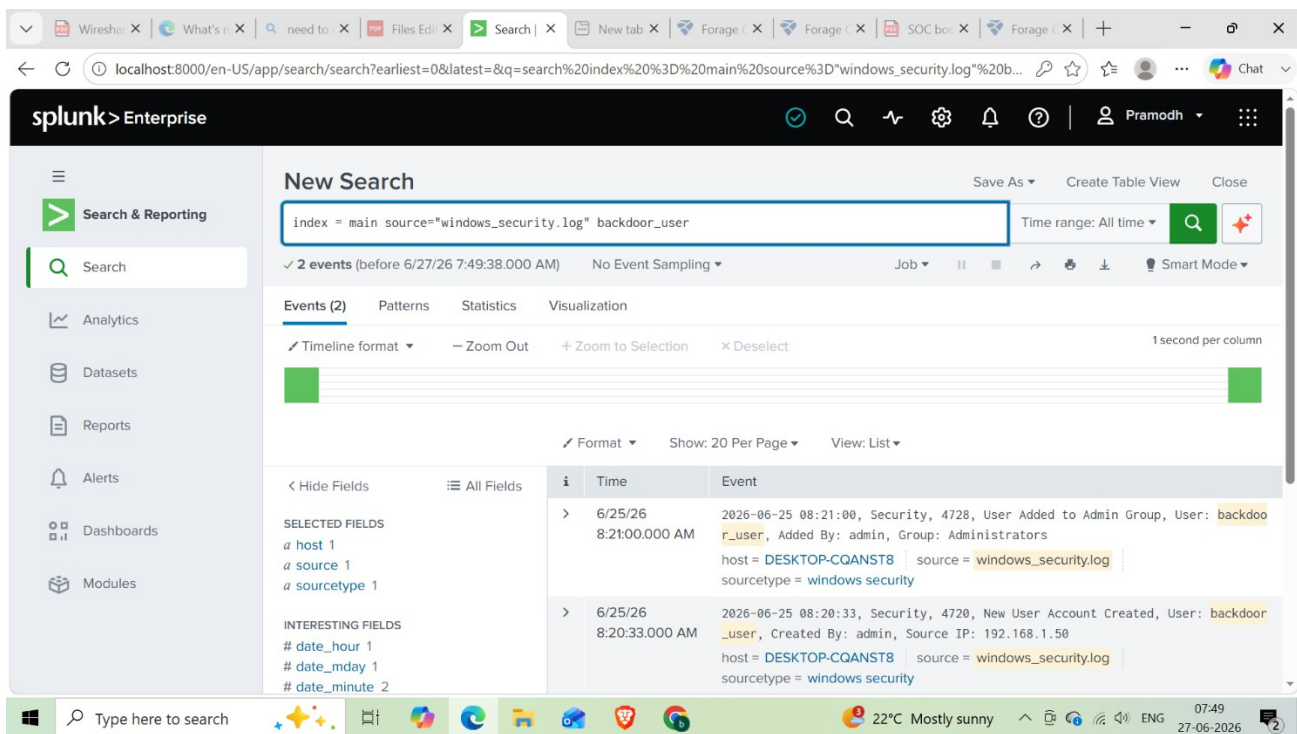


Figure 5 — backdoor_user account created (4720) and added to Administrators group (4728)

3.6 Persistence — Scheduled Task

At 08:45:00, a scheduled task was created with Event ID 4698. The task was named malware_persistence.exe in real attacks this would be disguised as a legitimate Windows process name like windows_update.exe to avoid detection. A scheduled task ensures that even if the machine is rebooted, the malware runs again automatically. Combined with the backdoor account, the attacker now has two separate persistence mechanisms.

The screenshot displays the Splunk Enterprise web interface. The search bar contains the query `index = main source="windows_security.log" 4698`. The search results show 1 event. The event details are as follows:

Time	Event
2026-06-25 08:45:00, Security, 4698, Scheduled Task Created, User: admin, Task: malware_persistence.exe, Source IP: 192.168.1.50	host = DESKTOP-CQANST8 source = windows_security.log sourcetype = windows security

Figure 6 — Scheduled task created at 08:45 for malware persistence

3.7 Defense Evasion — Audit Log Cleared

The very last action the attacker took was at 09:00:00 — clearing the Windows Security audit log using Event ID 1102. This is a defense evasion technique where the attacker tries to delete all evidence of what they did. However, because Splunk had already collected all the logs in real time before the clearing happened, the evidence was preserved in the SIEM and the full attack chain remained visible. This is exactly why forwarding logs to a SIEM in real time is critical an attacker cannot delete logs that have already been sent elsewhere.

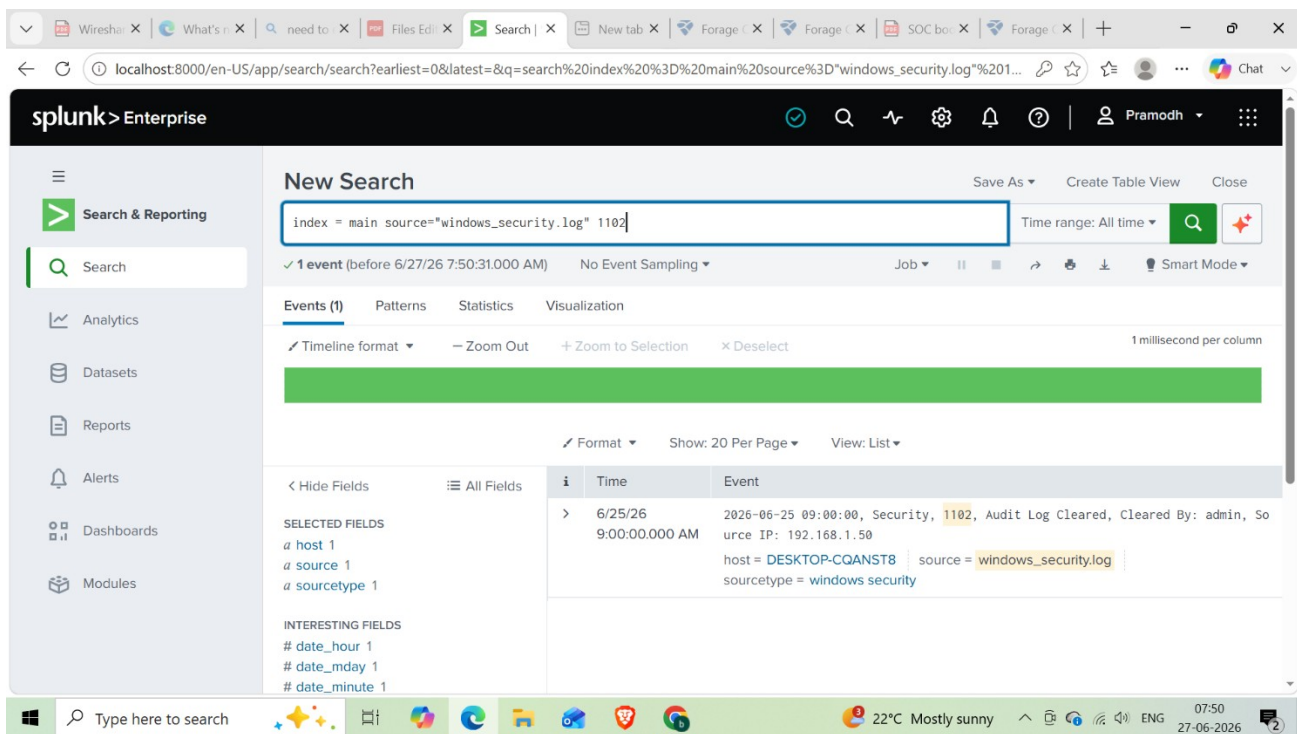


Figure 7 — Event 1102 — Audit log cleared by attacker at 09:00 AM

3.8 Attack Timeline — Investigation 1

Time	Event ID	What Happened	Severity
08:01:12	4624	john.doe logged in normally — legitimate user	Low
08:02:45	4624	jane.smith logged in normally — legitimate user	Low
08:10:01	4625	Failed login — admin — from 192.168.1.50 — brute force starts	High
08:10:03	4625	Failed login — admin — from 192.168.1.50	High
08:10:05	4625	Failed login — admin — from 192.168.1.50	High
08:10:07	4625	Failed login — admin — from 192.168.1.50	High
08:10:09	4625	Failed login — admin — from 192.168.1.50	High
08:10:11	4625	Failed login — admin — from 192.168.1.50	High
08:10:13	4625	Failed login — admin — from 192.168.1.50	High
08:10:15	4624	Successful login — admin — attacker got in via brute force	Critical
08:15:22	4688	cmd.exe launched — attacker setting up environment	High
08:15:25	4688	powershell.exe launched from cmd.exe	High
08:16:10	4688	mimikatz.exe launched — credential dumping from memory	Critical
08:20:33	4720	Backdoor_user account created by attacker	Critical
08:21:00	4728	backdoor_user added to Administrators group	Critical
08:45:00	4698	Scheduled task malware_persistence.exe created	High

09:00:00	1102	Audit log cleared — attacker covering tracks	Critical
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4. Investigation 2 — Ransomware and Insider Threat

This investigation covers a more complex incident involving four separate attack elements happening on the same day. An internal IT Admin performed network reconnaissance, an external attacker brute forced the james.wilson account via RDP and deployed ransomware, and at the same time a legitimate employee named emma.davis was secretly sending sensitive company files to an external IP address using FTP.

4.1 All Events Overview

The investigation3.log contained 43 events across a 2-hour window from 07:00 to 09:00 on June 26 2026. The timeline showed normal morning logins followed by suspicious scanning activity, then a rapid escalation of attack events all happening within the same timeframe as an insider threat.

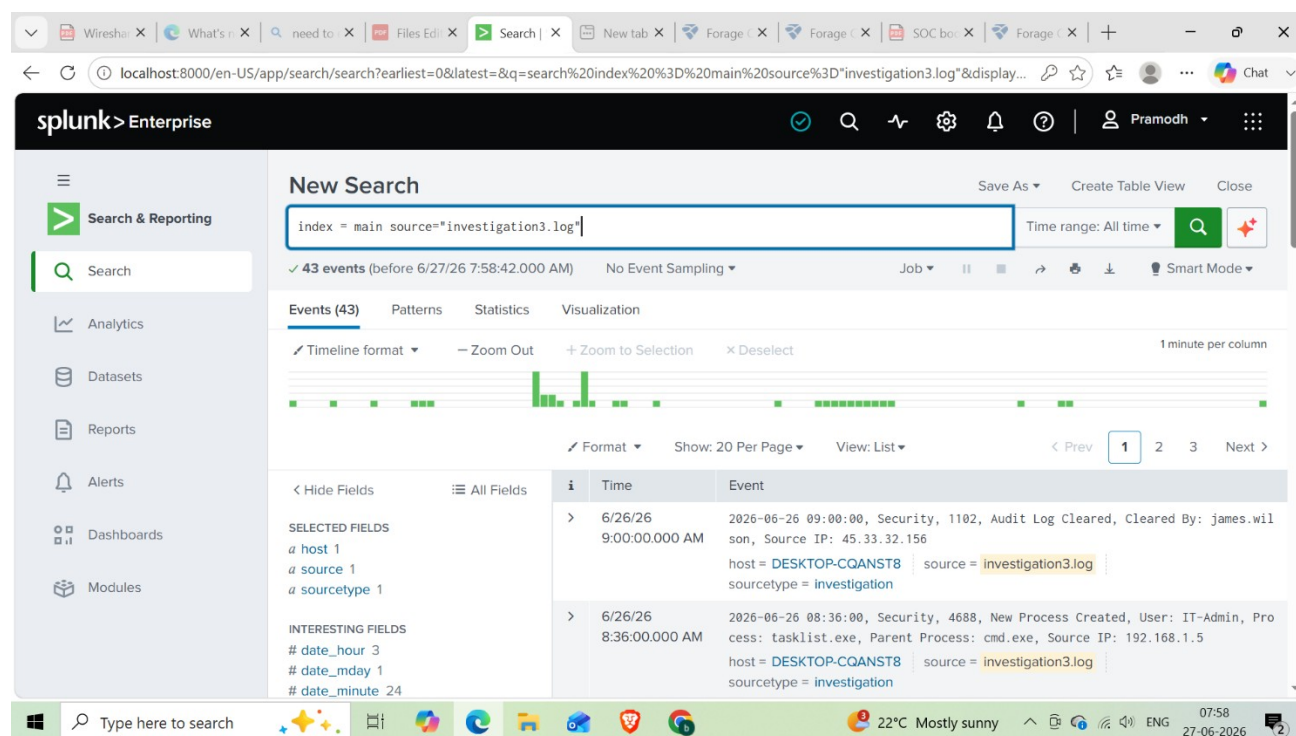


Figure 8 — All 43 events from investigation3.log showing activity timeline

4.2 Network Reconnaissance — Nmap Scan

At 07:15 to 07:17, IT-Admin from internal IP 192.168.1.5 ran nmap.exe three times scanning the entire internal subnet 192.168.1.0/24 across multiple ports including 22 (SSH), 80 (HTTP), 443 (HTTPS), 445 (SMB), and 3389 (RDP). While nmap is a legitimate network tool, scanning the entire internal network including RDP port 3389 is unusual behaviour. In a real SOC investigation, the analyst would contact IT-Admin to verify if this scan was

authorized. If it was not authorized, this could indicate an insider doing reconnaissance to identify targets for lateral movement.

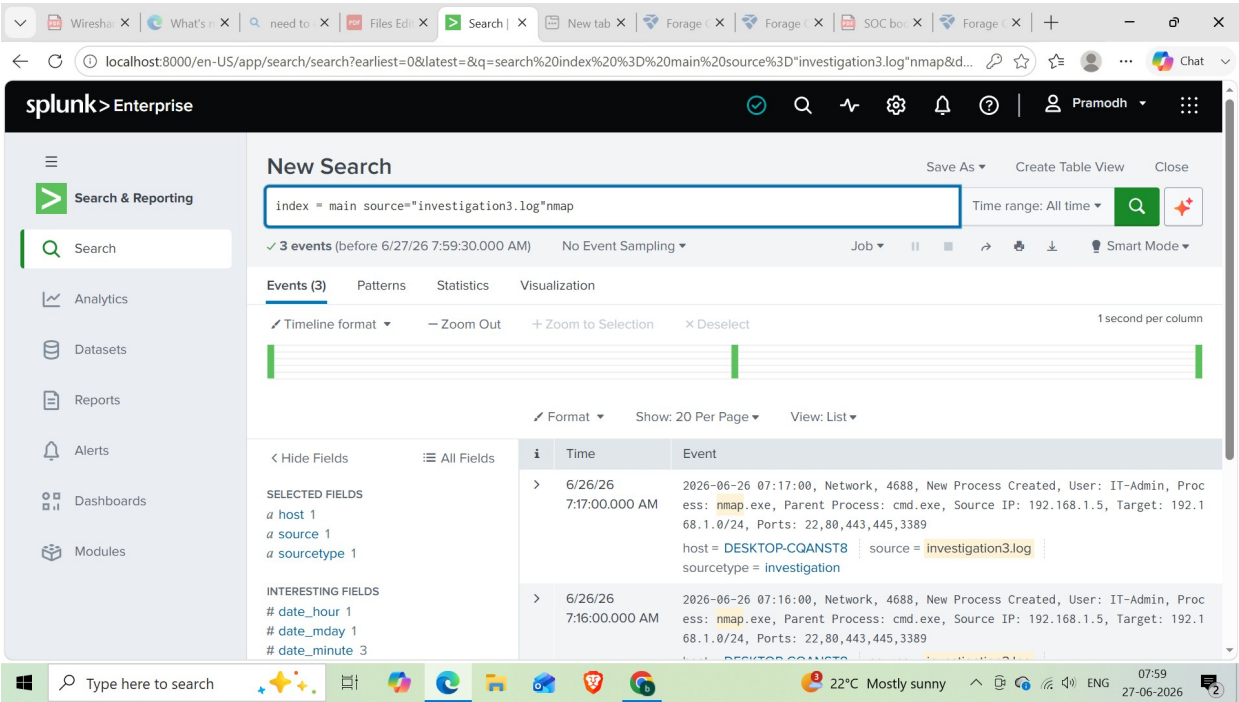


Figure 9 — Nmap scan by IT-Admin across internal subnet on multiple ports

4.3 Brute Force — james.wilson Account

At 07:30, an external attacker from IP 45.33.32.156 launched a brute force attack against the james.wilson account using RDP — Logon Type 10. Five failed login attempts were recorded in rapid succession between 07:30:00 and 07:30:08, followed by a successful login at 07:30:10. The entire brute force took just 10 seconds, indicating an automated tool was used. This is particularly suspicious because james.wilson had already logged in normally from his internal IP 192.168.1.40 at 07:00 — meaning the real james.wilson was already at work when the attacker was trying to access his account from outside.

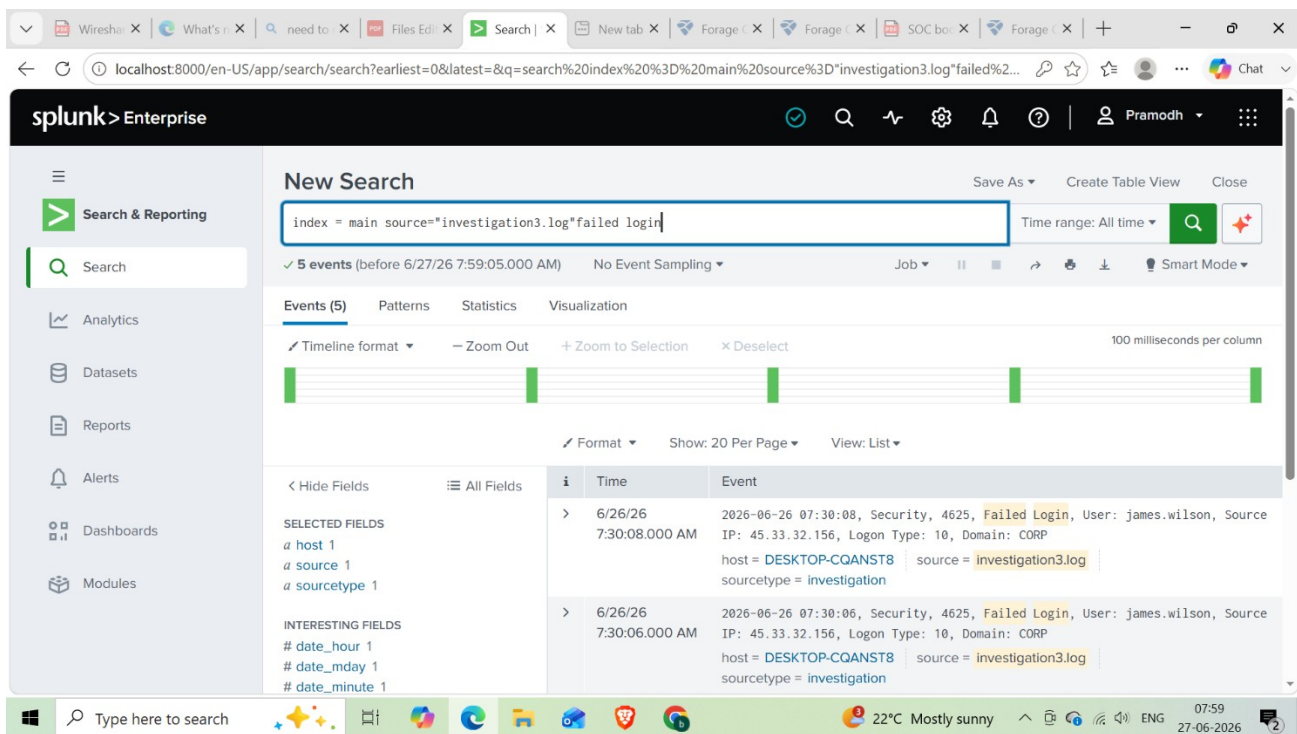


Figure 10 — 5 failed login attempts on james.wilson from external IP 45.33.32.156

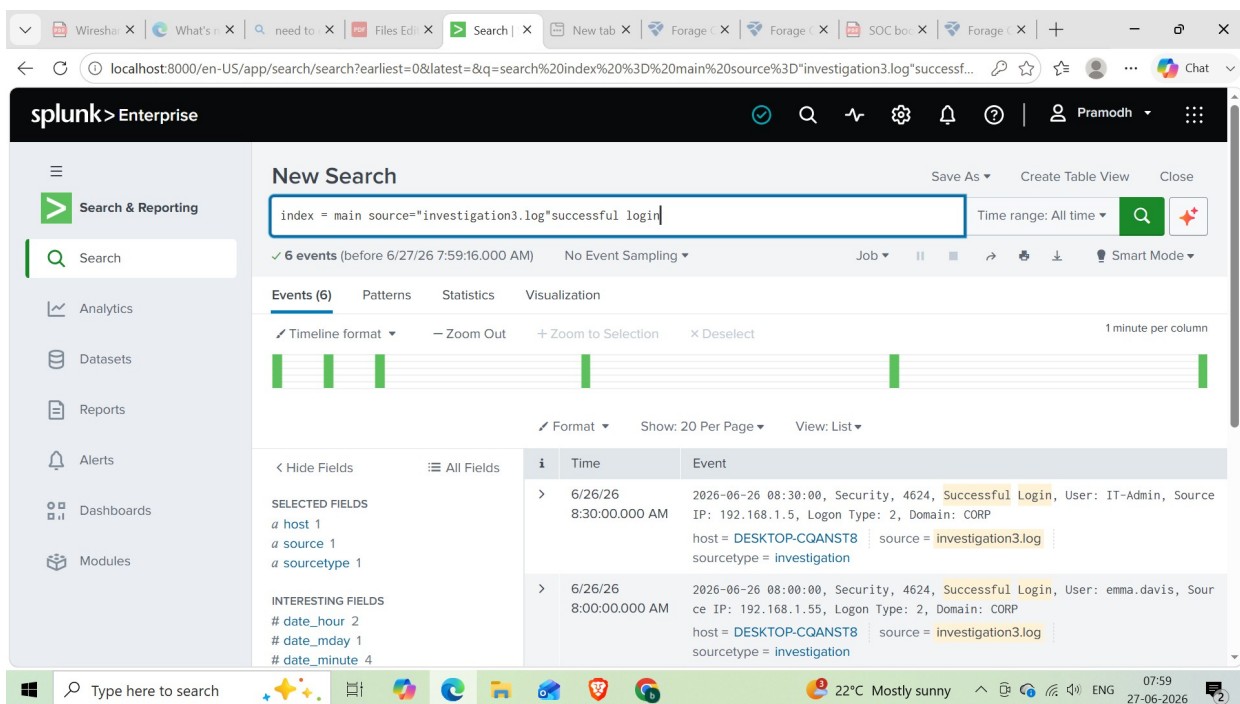


Figure 11 — Successful login by attacker using james.wilson account via RDP

4.4 Full Attacker Activity — IP 45.33.32.156

Searching for all events from the attacker IP 45.33.32.156 revealed 22 events showing the complete post-compromise activity. After getting into the james.wilson account, the attacker immediately started a discovery phase running whoami, ipconfig, and net commands to understand the environment they were in. They then escalated to credential theft and ransomware deployment.

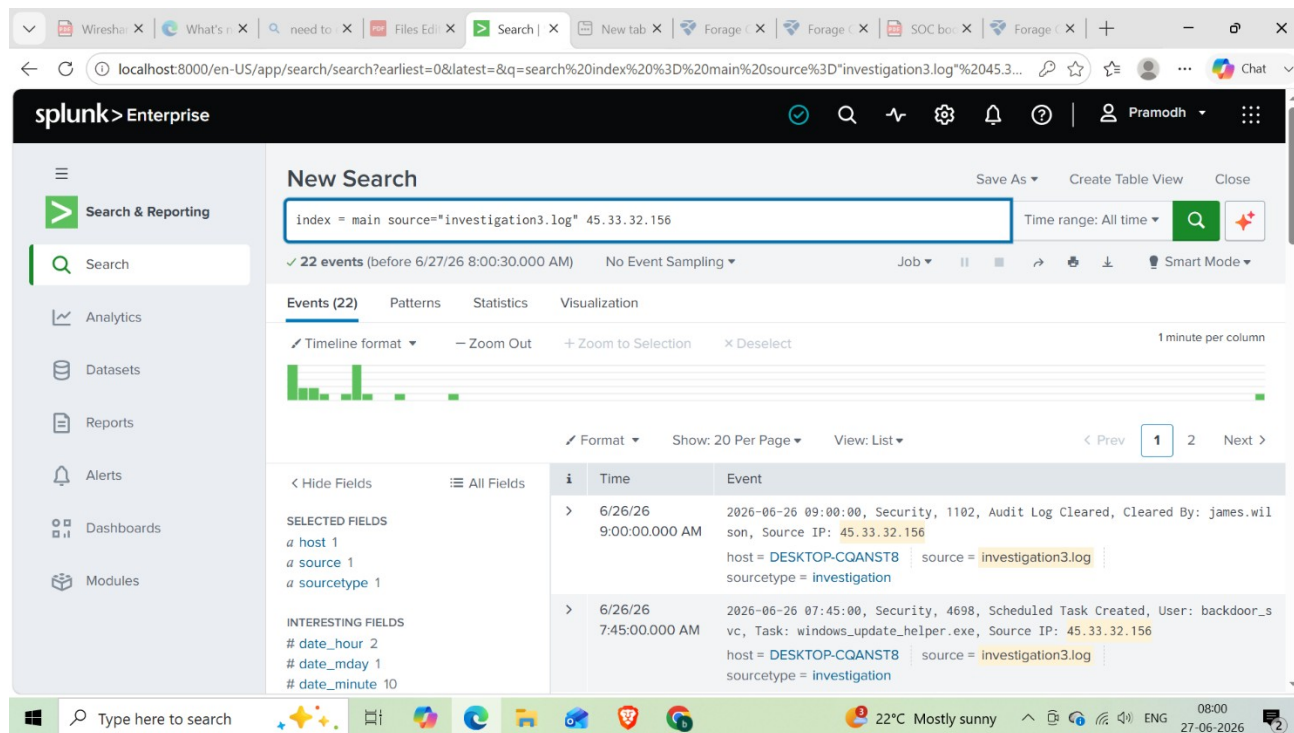


Figure 12 — All 22 events from attacker IP 45.33.32.156 showing full attack chain

4.5 Ransomware Execution

At 07:36:00, ransomware_payload.exe was launched from powershell.exe under the james.wilson account. One minute later at 07:37:00, notepad.exe was launched by ransomware_payload.exe to display the ransom note to the victim a file called README_RANSOM.txt. This is standard ransomware behavior encrypt files first, then display the ransom demand asking for payment to get the files back.

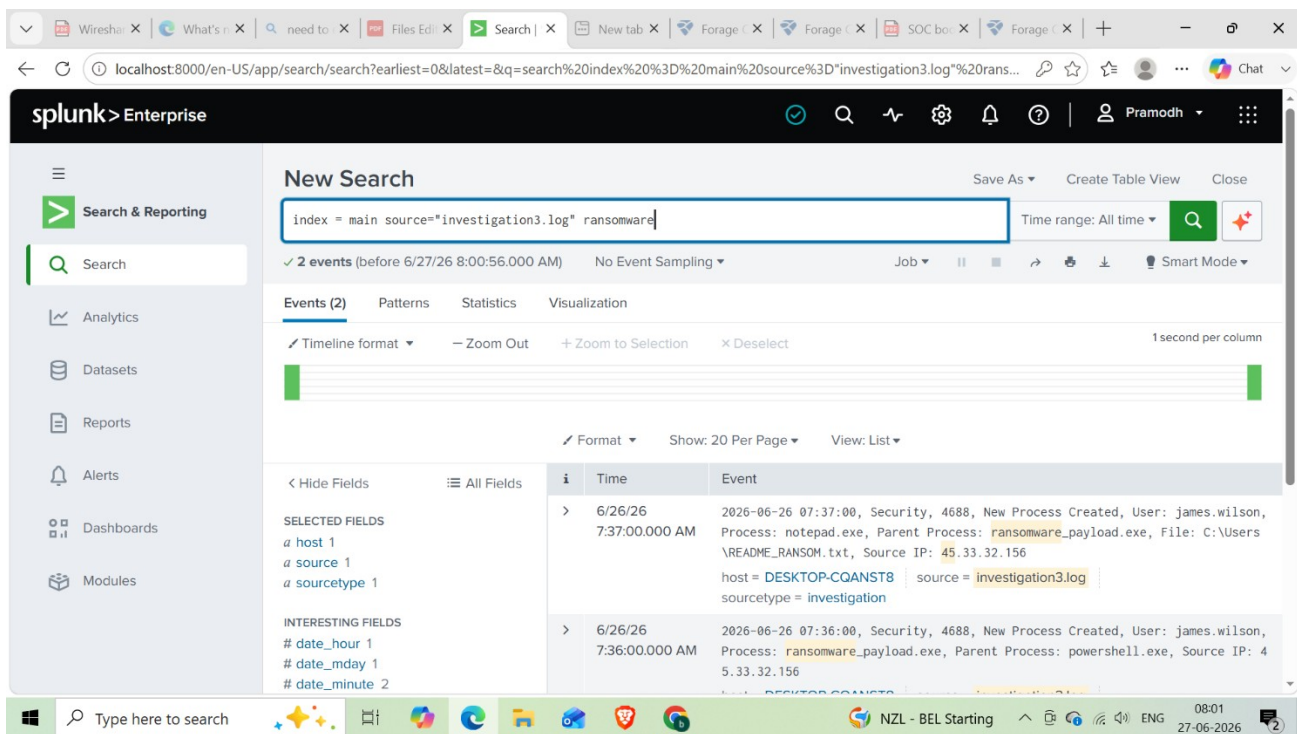


Figure 13 — ransomware_payload.exe executed and notepad.exe launched for ransom note

4.6 Files Encrypted by Ransomware

The ransomware encrypted 5 critical files across different sensitive directories. All file access events used Event ID 4663 and all files have the encrypted extension added by the ransomware. The files targeted were spread across Finance, HR, Legal, and personal user folders showing the ransomware had access to the entire file system.

- C:\Users\james.wilson\Documents\project.docx.encrypted
- C:\Users\emma.davis\Documents\salary.xlsx.encrypted
- C:\Finance\annual_report.xlsx.encrypted
- C:\HR\employee_records.xlsx.encrypted
- C:\Legal\contracts.docx.encrypted

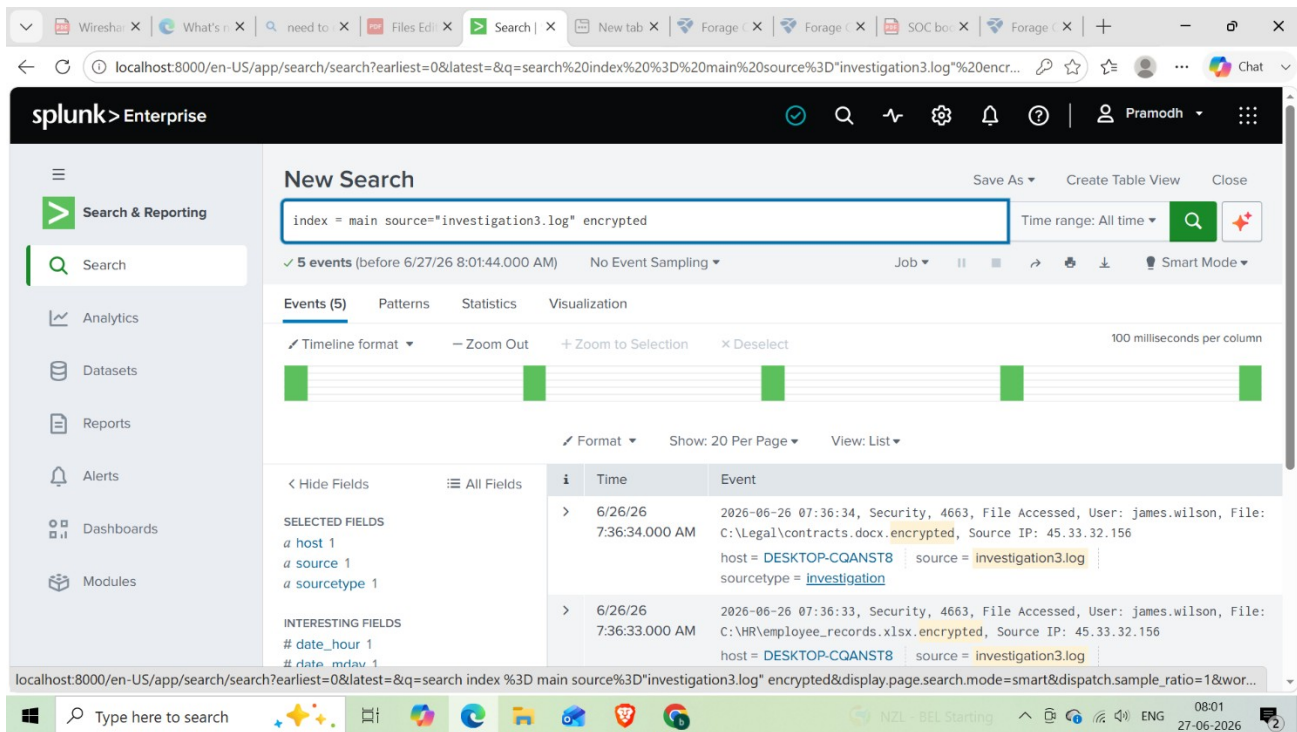


Figure 14 — 5 encrypted files showing ransomware impact across Finance, HR and Legal folders

4.7 Insider Threat — emma.davis Data Exfiltration

At 08:05 to 08:14, a separate threat was identified running in parallel with the external attack. emma.davis, a legitimate employee logged in from her internal IP 192.168.1.55, accessed 5 sensitive files across Finance, HR, and Legal folders using Event ID 4663. She then launched ftp.exe and transferred 4 of those files to an external IP 203.0.113.50. The total data transferred was approximately 10.3 MB of sensitive company information including financial reports, employee records, acquisition plans, and salary data. This is a classic insider threat a trusted employee abusing their access to steal company data.

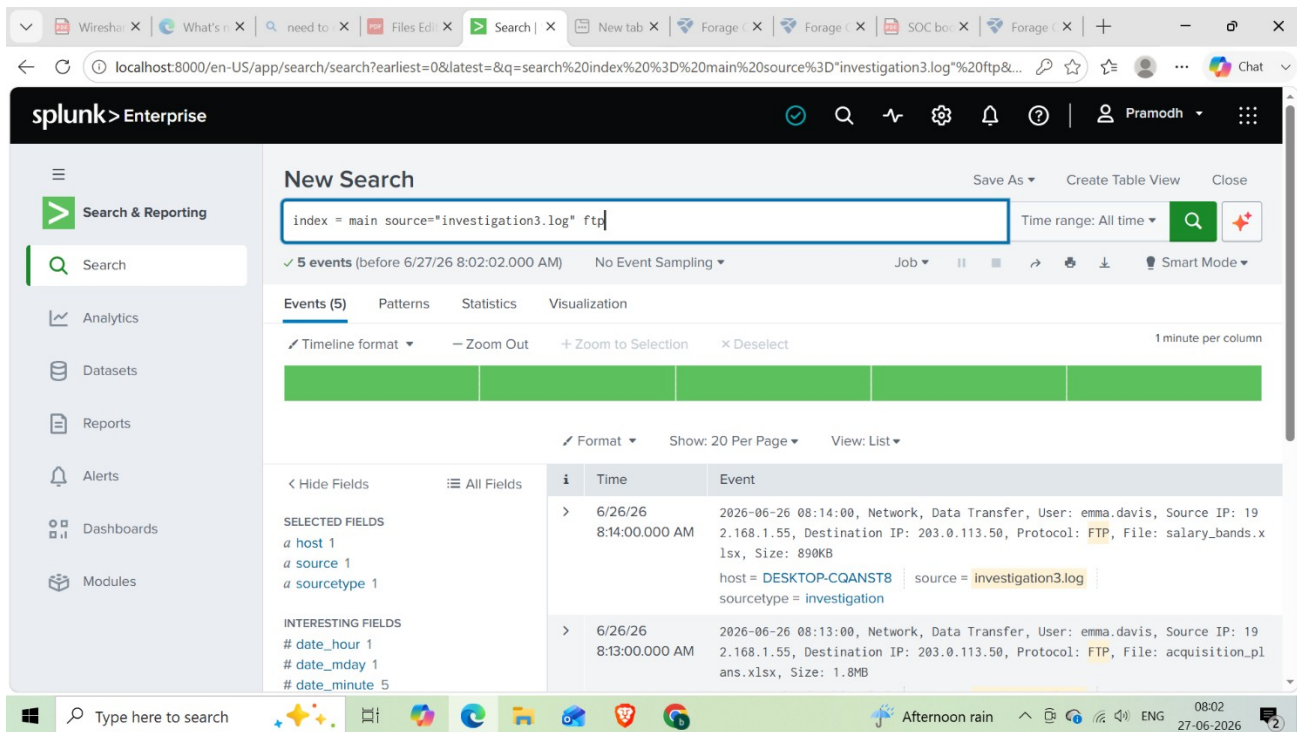


Figure 15 — emma.davis using FTP to transfer sensitive files to external IP 203.0.113.50

4.8 Defense Evasion — Audit Log Cleared

At 09:00:00, the attacker using the james.wilson account from IP 45.33.32.156 cleared the Windows audit log using Event ID 1102. As with Investigation 1, this attempt to destroy evidence was unsuccessful because Splunk had already collected all the logs in real time. The attacker cleared the local Windows logs but the evidence was already preserved in the SIEM.

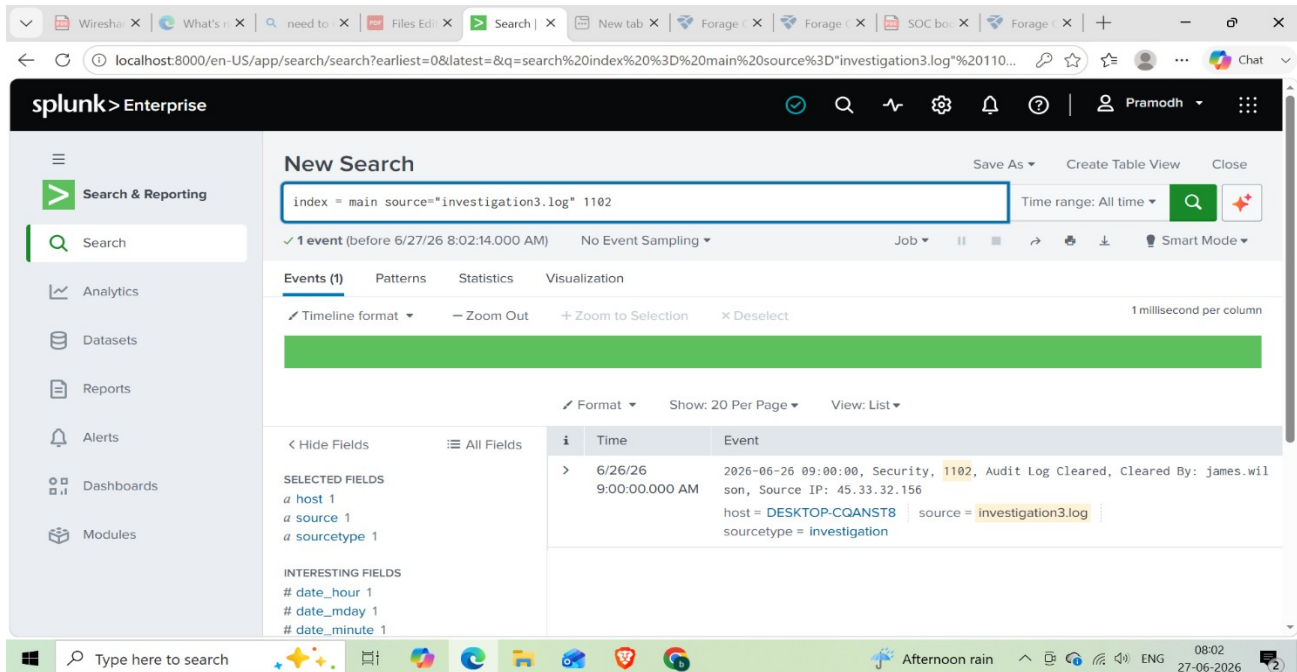


Figure 16 — Event 1102 Audit log cleared by attacker at 09:00 AM

4.9 Attack Timeline — Investigation 2

Time	Event ID	What Happened	Severity
07:00:01	4624	james.wilson normal login — internal IP 192.168.1.40	Low
07:05:00	4624	emma.davis normal login — internal IP 192.168.1.55	Low
07:10:00	4624	IT-Admin login — internal IP 192.168.1.5	Low
07:15-17	4688	IT-Admin runs nmap.exe — scans entire internal subnet	Medium
07:30:00	4625	Brute force starts on james.wilson from 45.33.32.156 via RDP	High
07:30:10	4624	Brute force succeeded — attacker inside james.wilson account	Critical
07:31:00	4688	cmd.exe and powershell.exe launched by attacker	High
07:32:00	4688	whoami, ipconfig, net.exe — attacker doing discovery	High
07:36:00	4688	ransomware_payload.exe executed from powershell	Critical
07:36:30	4663	5 files encrypted — Finance, HR, Legal, user folders	Critical
07:37:00	4688	notepad.exe opens README_RANSOM.txt — ransom demand	Critical
07:40:00	4720	backdoor_svc account created by attacker	Critical
07:41:00	4728	backdoor_svc added to Administrators group	Critical
07:45:00	4698	Scheduled task windows_update_helper.exe created	High
08:05-09	4663	emma.davis accesses Finance HR and Legal files	High
08:10:00	4688	emma.davis launches ftp.exe — data exfiltration begins	Critical
08:11-14	Transfer	4 files totaling 10.3 MB sent to 203.0.113.50 via FTP	Critical
09:00:00	1102	Audit log cleared by attacker — defense evasion	Critical

5. MITRE ATT&CK Mapping

All attack techniques identified in both investigations have been mapped to the MITRE ATT&CK framework. This framework provides a standardized language for describing attacker behavior and is used by SOC teams worldwide to communicate findings clearly and consistently.

Tactic	Technique	ID	Evidence in Investigation
Initial Access	Brute Force — Password Guessing	T1110	7 failed logins then successful admin login from 192.168.1.50
Initial Access	Brute Force via RDP	T1110	5 failed RDP logins on james.wilson from 45.33.32.156
Execution	Command and Scripting — PowerShell	T1059.001	cmd.exe spawning powershell.exe after compromise
Discovery	System Information — whoami ipconfig	T1082	Attacker ran whoami, ipconfig, net.exe after getting in
Credential Access	OS Credential Dumping — Mimikatz	T1003	mimikatz.exe launched from powershell.exe at 08:16
Persistence	Create Account — Local Account	T1136.001	backdoor_user and backdoor_svc accounts created
Persistence	Account Manipulation — Admin Group	T1098	Both backdoor accounts added to Administrators
Persistence	Scheduled Task	T1053.005	malware_persistence.exe and windows_update_helper.exe tasks
Defense Evasion	Indicator Removal — Clear Logs	T1070.001	Event 1102 — audit log cleared in both incidents
Exfiltration	Exfiltration Over Alternative Protocol — FTP	T1048	emma.davis sent 10.3 MB to 203.0.113.50 via FTP
Impact	Data Encrypted for Impact — Ransomware	T1486	ransomware_payload.exe encrypted 5 critical files

6. Containment Actions

6.1 Immediate Actions — Do Right Now

- Block IP 192.168.1.50 on the firewall — this is the brute force attacker from Investigation 1
- Block IP 45.33.32.156 on the firewall — this is the ransomware attacker from Investigation 2
- Block IP 203.0.113.50 on the firewall — this is where emma.davis sent the stolen data
- Isolate DESKTOP-CQANST8 from the network immediately — stop any further damage
- Disable backdoor_user account — created by attacker in Investigation 1
- Disable backdoor_svc account — created by attacker in Investigation 2
- Suspend emma.davis account pending HR and legal investigation
- Reset admin account password immediately
- Reset james.wilson account password immediately

6.2 Short Term Actions — Within 24 Hours

- Delete scheduled task malware_persistence.exe from DESKTOP-CQANST8
- Delete scheduled task windows_update_helper.exe from DESKTOP-CQANST8
- Remove backdoor_user and backdoor_svc from Administrators group
- Restore the 5 encrypted files from the most recent clean backup
- Check all other machines on the network for signs of lateral movement using the stolen mimikatz credentials
- Geolocate IP 45.33.32.156 and 203.0.113.50 identify which country they are in
- Check VirusTotal for reputation of both external IPs
- Reset passwords for all user accounts mimikatz may have stolen all of them
- Investigate emma.davis involve HR and legal team immediately
- Determine what data was in the 4 files sent to 203.0.113.50 and assess regulatory impact

6.3 Long Term Actions — Within 1 Week

- Implement account lockout policy lock account after 3 failed login attempts
- Enable Multi Factor Authentication on all RDP connections
- Block FTP outbound traffic on the firewall use SFTP instead if file transfer is needed
- Deploy EDR on all endpoints to detect mimikatz and ransomware execution in real time
- Set up real time Splunk alerts for brute force patterns 5 or more failed logins in 1 minute
- Set up alert for Event ID 4720 and 4728 any new account creation triggers immediate alert
- Set up alert for Event ID 1102 any log clearing triggers critical alert
- Review all user access permissions apply least privilege principle
- Conduct security awareness training for all employees on insider threat and phishing

7. Recommendations

Based on the findings from both investigations, the following recommendations are made to prevent similar incidents from happening again.

Priority	Recommendation	Why it helps
Critical	Enable MFA on all RDP connections	Even with correct password attacker cannot login without second factor
Critical	Implement account lockout after 3 failed attempts	Stops brute force before attacker can find correct password
High	Deploy EDR on all endpoints	Detects mimikatz and ransomware execution in real time before damage
High	Block outbound FTP and restrict data transfer tools	Prevents insider threats from easily sending data outside the network
Medium	Apply least privilege — limit who can access Finance and HR files	Limits damage if any account is compromised or misused
Medium	Implement regular offline backups of all critical files	Ransomware loses its power if clean backups are available for recovery

8. Conclusion

This investigation demonstrated how Splunk SIEM can be used to detect, investigate, and document complex cyberattacks in a structured way. Two separate but equally serious incidents were uncovered from log analysis alone an external brute force attack leading to ransomware deployment, and a parallel insider threat involving unauthorized data exfiltration.

The most important finding from both investigations is that the attackers tried to delete their tracks by clearing the Windows audit logs using Event ID 1102. However, because Splunk was collecting logs in real time and storing them in the SIEM, this attempt to cover their tracks completely failed. All evidence was preserved and the full attack chains were reconstructed from start to finish.

From a SOC analyst perspective, this exercise reinforced the critical importance of real time log forwarding to a SIEM, setting up detection rules for known attack patterns like brute force and new account creation, and having a clear incident response process that moves quickly from detection to containment. The MITRE ATT&CK framework provided a common language to describe every technique the attackers used, which makes communication with the wider security team and management much faster and clearer.

Disclaimer

The log files used in this investigation were simulated using real Windows Security Event IDs (4624, 4625, 4688, 4720, 4728, 4698, 1102) and real attack patterns documented in the MITRE ATT&CK framework. The scenarios are based on real world attack techniques observed in actual SOC environments. This project was created for educational and portfolio purposes to demonstrate SOC analyst investigation skills using Splunk SIEM.

--- End of Report ---